F.CSM302

1

The Sleeping Teaching Assistant

Thread, Semaphore, Mutex synchronization
Tkinter

: **II-4-10**

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GitHub: https://github.com/B242270001/II-4-10_SleepingTeachingAssistant

† *f* , 2026

€

• 4

1. , ... 5

1.1 ‰ … ƒ 5

1.2 ƒ … 5

1.3 Š , ‹ 5

1.4 ‰ ˆ … ƒ ˆ ƒ € … 5

1.5 … ƒ 7

10. 8

10.1 … € 8

10.2 8

10.3 ƒ ‹ 13

10.4 …€ ƒ 14

10.5 , … 14

ƒ „ … † 15

‡ ˆ „

Š 1. ‰ ˆ … ƒ ˆ ƒˆ 6

• „ ˆ „

1. Core- … synchronization •ˆ 6

2. … ƒ 7

3. 8

4. 8

5. ˆ … ‹ 9

6. … 11 ˆ … … 9

7. Edge case 27 ˆ … … (tests/test_edge_cases.py), ... 10

8. UI ˆ *f* *f* 11

9. Core (UI- …) € *f* € 12

10. UI *f* core- … *f* Ž 12

11. UI ˆ *f f* € 13

•

, ƒ ˆ Silberschatz- … *Operating System Concepts* (10th edition) ƒ Œ … 7- … •The Sleeping Teaching Assistant• ‘ Œ … Python ƒ ƒ € . … ƒ (TA) ƒ ‹ ƒ … , ’ . • ’ …ƒ … ˆ TA €, ’ ƒ . ‰ ˆ ’ .

… ƒ “ ƒ ƒ . **Core** (`ta_core.py`) Ž TA ’ … thread € €, ƒ … Lock (mutex)- ƒ €, ƒ ƒ … Semaphore- ƒ € . Core Ž UI- ƒ … thread … … … event (queue.Queue) ƒ … . **UI** (`ui.py`) Ž Tkinter dashboard event- … € TA, , ’ ƒ , , ” , event ƒ ƒ . Œ , ’ , ƒ … Œ … ƒ ƒ €, START, PAUSE, STEP, RESET … ƒ .

ƒ … : … 11 ˆ , 27 edge case ˆ , 11 UI ˆ . ƒ ˆ . Synchronization- € ƒ FIFO , ƒ ƒ … …ƒ, TA ’ … Œ ƒ, deadlock- … …ƒ Ž ƒ ƒ , … Œ (20 , 50 ’) ˆ . Ž Œ ƒ … ˆ UI- … core- ƒ ƒ ’ .

‰ **:** synchronization, thread, semaphore, mutex, Sleeping Teaching Assistant, Python, Tkinter.

1. ,

1.1 Š ‰ ‹

• Œ ˆ … … … ˆ ‘ ˆ , thread- … ‹ € . ,
Ž … … (ƒ …, , ƒ €)- … ƒ ƒ ˆ ƒ Œ
race condition, deadlock ‹ … . … mutex
semaphore- … ‹ ƒ €, thread- … ‹ € ƒ Ž
… … ˆ … … ’ . Sleeping Teaching Assistant
Ž … … € … ƒ ƒ
€ Œ ƒ .

1.2 ƒ„ ‰

• ƒ ƒ … …:

• TA- … (TA), N ƒ … … .

• • ’ …ƒ … TA .

• • ’ ƒ TA € … . TA ‹ …
ƒ . ‰ …ƒ … € , ƒ ƒ ƒ .

• • ’ FIFO … Œ .

€ ƒ “ … … ƒ …. • ,
ƒ … TA- … … thread ‹ Œ ƒ (race
condition). “ , TA ’ ƒ € ƒ, ’ TA-
€ Œ … ƒ (lost wakeup, deadlock) … .

1.3 ‡ , Œ

‡ : Sleeping Teaching Assistant thread, semaphore, mutex ‹
ƒ € €, € Ž ƒ ƒ … ƒ.

‡ :

• UI- ƒ …, € ƒ … core Œ ƒ (`ta_core.py`).

• Core- … thread- … … ƒ ˆ Tkinter dashboard
ƒ …ƒ (`ui.py`).

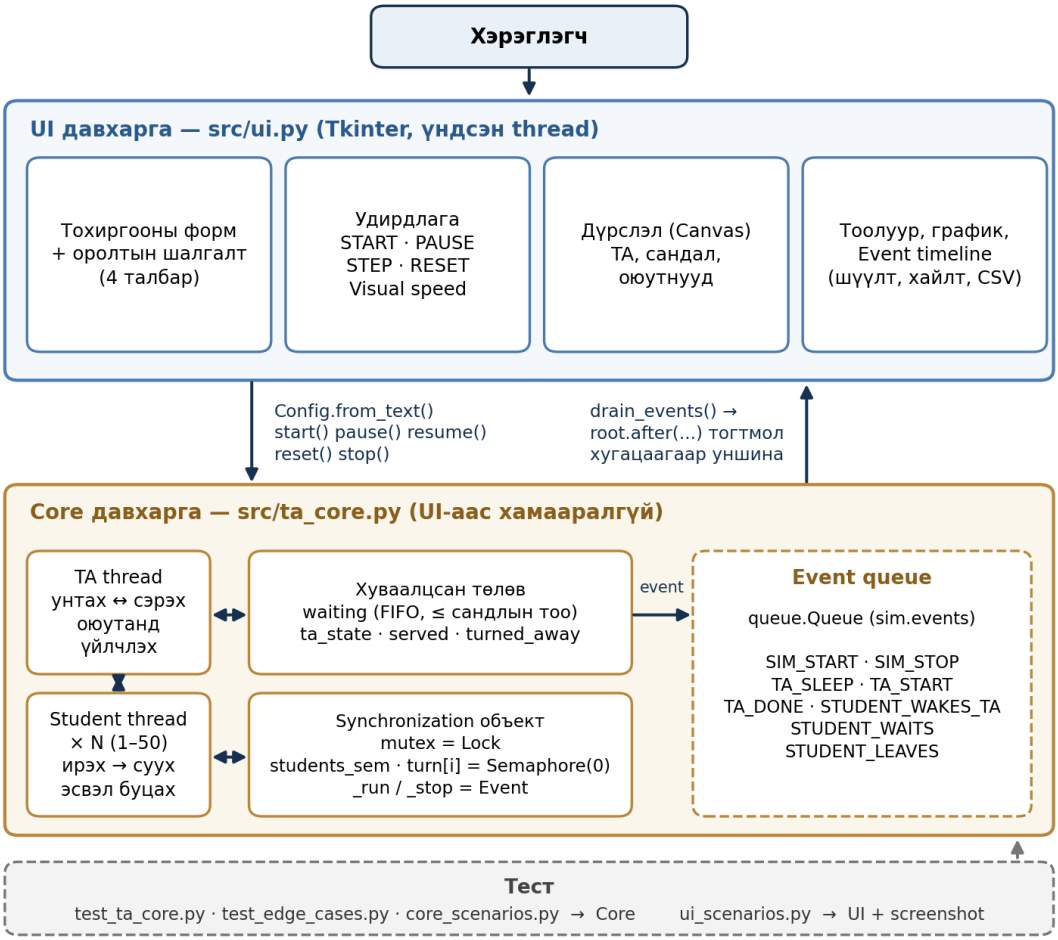
• • €, … € … ˆ ˆ€ ƒ.

• START, PAUSE, STEP, RESET , event ƒ … ƒ, ƒ …ƒ, CSV
ƒ ƒ € ƒ.

• ˆ , edge case ˆ , UI ˆ ‹ … € ƒ.

1.4 „ †… ‰ † ‹ ‰ ˆ „

– … Ž ‹ … € . ‰ ˆ Ž ƒ “ ƒ : UI-
ƒ … **Core** Tkinter **UI**. Thread … … … (’ , ,
, TA , … Œ € ƒ , ,) event queue.Queue-
ƒ … . Tkinter- … thread Ž `root.after(...)`- €
‹ . … Ž core- … € ƒ €, UI
thread Ž core- … ƒ ƒ ….



1.

• „ 1. Core- synchronization ‹•†

‹•†	Ž Ž	
mutex	threading.Lock	… , TA- … , *f*
students_sem	Semaphore(0)	€ … ’ . TA *wait f* …€ , ’ *signal* Œ
turn[i]	Semaphore(0)	i- ’ *f* … semaphore. TA … Œ € ’ *f*
_run	threading.Event	Set = € € … , clear = ‹ (pause)
_stop	threading.Event	*f* thread- … ‹ *f* *f*

1.5 Ž„ ‹ • ‹ ‰ „

• „ 2. Ž„ ‹ • ‹ ‰ „

‡	*f*
`src/ta_core.py`	Core: *Config*, *Simulation*, thread- ,
`src/ui.py`	Tkinter dashboard
`src/test_ta_core.py`	… 11 ˆ (*unittest*)
`tests/`	27 edge case ˆ , core UI ˆ … ‘ , screenshot
`docs/`	Report, Manual
,	Python 3.14, Tk 9.0. ‰ : *threading*, *queue*, *tkinter*, *unittest*
	Git, GitHub (https://github.com/B242270001/II-4-10_SleepingTeachingAssistant)

10.

10.1 • „ ˆ

Sleeping Teaching Assistant thread, mutex, semaphore
ƒ € €, UI- ƒ … core Tkinter dashboard- . Dashboard
Ž € , … START, PAUSE, STEP, RESET- ƒ, TA
’ ƒ … , ” … ƒ ƒ, event
ƒ … €, ƒ …€, CSV ƒ ƒ € …. Œ …
operational (Manual) .

10.2

ƒ … . ƒ ƒ Œ ,
, ˆ ˆ … ƒ € ƒ … . ‰ …
€ ….

• „ **3.** **€**

‡	,
–… … ˆ	macOS 26.6.2
Python / Tk	3.14.10 / Tk 9.0
… ƒ	main, commit 198e554 (release- … … commit/tag- Œ)
Core	src/ta_core.py (thread, Semaphore, Lock)
UI	src/ui.py (Tkinter dashboard), Visual speed 1.00˜
•	2026.10.28

• „ **4.** „

Š	•	†„		• •
… ˆ	src/test_ta_core.py	11	11 / 11	14.97 ˆ
Edge case ˆ	tests/test_edge_cases.py	27	27 / 27 (*OK*)	22.8 ˆ
Core ˆ	tests/core_scenarios.py	6 ˆ (S1™S6)	JSON- ƒ	~2
UI ˆ	tests/ui_scenarios.py	11 ˆ (S1™S11)	11 / 11	~3.5
		38 Š … †„ + 11 UI „•†	•„	

• „ 5. ƒŠ … †„ Œ

•	†„	†„	‘
test_ta_core.py	TestValidation	2	(8 ƒ), ‹ ƒ ‹
test_ta_core.py	TestCorrectness	6	Invariant (, ƒ , … Œ), Œ , 0 , ƒ ƒ, deadlock
test_ta_core.py	TestControl	3	Pause / resume / stop, reset, ƒ start
test_edge_cases.py	TestConfigBoundaries	9	‹ , nan/inf, ƒ ‹ …, ƒ , seed, ˆ ˆ€
test_edge_cases.py	TestLifecycleEdgeCases	7	Stop, pause, reset- … ˆ , thread €
test_edge_cases.py	TestSimulationSemantics	11	FIFO, , ƒ , TA- … , Œ
		38	

• „ 6. 11 Š … †„

	†„	/ Ž‰•Ž	• ‰
1	test_bad_inputs	8 ƒ (-1, 21, "abc", 0 š)	ValueError
2	test_good_and_edge_inputs	3/5/1.5/2; 0/1/0.01/0.01; 20/50/60/60	…
3	test_book_default	3 , 5 ’ , 2 ˆ	Invariant ˆ , … Œ > 0, thread 0
4	test_overload_students_get_turned_away	1 , 20 ’	> 0, STUDENT_LEAVES
5	test_zero_chairs	0 , 5 ’	–… Œ 0, > 0, ƒ Œ ƒ …
6	test_ta_sleeps_when_nobody_comes	ƒ ƒ 30 ˆ	TA , TA_START ƒ …
7	test_student_wakes_ta	1 ’	STUDENT_WAKES_TA
8	test_no_deadlock_many_configs	1/3, 2/10, 5/30, 3/3	–… Œ > 3, thread 0, invariant ˆ
9	test_pause_resume_stop	pause › resume › pause › stop	Pause ˆ event 0, resume € , ƒ …
10	test_reset_with_new_config	reset(Config)	, event 0, ƒ €
11	test_double_start_rejected	start() ƒ “	RuntimeError

• „ **7. Edge case 27 †„ (tests/test_edge_cases.py), ‹ •„**

		Edge case	**/ Ž‰•Ž**	• ‰
1	‹	‰ ƒ ‹	0, 20 / 21, -1	0, 20 ‹ . 21, -1
2	‹	• ’ ƒ ‹	1, 50 / 0, 51	1, 50 ‹ . 0, 51
3	‹	ƒ ‹	0.001, 60 / 0, 60.01	0.001, 60 ‹ . 0, 60.01
4	‹	nan, inf, -inf		ValueError
5	‹	‹ …, 1e1	" 3 ", "1e1"	Š (3, 10.0)
6	‹	ƒ ƒ …, ˆ , ƒ	"3.0", "5.5", "", " "	ValueError
7	‹	Seed	"", "-5" / "abc", "1.5"	None, -5 ‹ .
8	‹	ˆ ˆ€	"abc"	"‰ ", "abc" … € … ˆ ˆ€
9	‹	Simulation-Config	num_chairs=-1	ValueError
10	Lifecycle	, ƒ ƒ stop	stop()	ƒ …
11	Lifecycle	“ stop	stop(); stop()	ƒ …, thread 0
12	Lifecycle	, ƒ ƒ pause	pause(); start()	Resume ƒ ‹ ƒ SIM_START
13	Lifecycle	Pause ˆ reset	pause(); reset()	Pause , event
14	Lifecycle	ƒ … reset	reset(Config(99))	ValueError, ‹ . Ž ƒ … reset
15	Lifecycle	6 reset	start › reset ˜ 6	Thread € ƒ …
16	Lifecycle	Event- …	stop	SIM_STOP , ƒ …
17	‰ ƒ Œ	FIFO	5 , 8 ’	, ƒ € Ž ƒ € … Œ
18	‰ ƒ Œ	ƒ Œ	2 , 10 ’	. LEAVES = turned_away
19	‰ ƒ Œ	–… Œ œ	3 , 6 ’	served ƒ ƒ …
20	‰ ƒ Œ	ƒ	3 , 10 ’	Event- … t ƒ …
21	‰ ƒ Œ	current	3 , 5 ’	TA_START- current = ’ , TA_DONE- None
22	‰ ƒ Œ	, ƒ … ’ TA-	1 ’	SIM_START › WAKES_TA › WAITS › TA_START

		Edge case	/ Ž‰•Ž	• ‰
23	‰ ƒ Œ	ƒ TA	1 ’ ,	TA_DONE- … TA_SLEEP
24	‰ ƒ Œ	1 , 1 ’	1 / 1	3- … Œ , ƒ ‹ Œ ƒ …
25	‰ ƒ Œ	‰ ƒ …	20 , 10 ’	Œ ƒ …
26	‰ ƒ Œ	Œ	20 , 50 ’	œ 20, ƒ ƒ …, thread 0, deadlock- …
27	‰ ƒ Œ	0	0 , 5 ’	TA ƒ , ƒ Œ … Œ ƒ …

• „ 8. UI „•† Ž ŽŠ Ž„Ž ‹ ‰ ‰

ID		Ž Ž		• ‰
S1	– €	…	3 / 5 / 1.0 / 1.5	TA ƒ . • ’ ƒ €, … Œ
S2	‰	‹	1 / 10 / 0.3 / 1.5	• ’ . ƒ ƒ …
S3	‰ ƒ …	…	10 / 3 / 1.5 / 0.5	Œ ƒ …, 0
S4	0	‹	0 / 5 / 0.5 / 1.0	Œ … Œ ƒ …, TA ƒ
S5	• ’	‹	1 / 1 / 2.0 / 0.5	• ’ TA €, Ž ƒ
S6	Œ	‹	20 / 50 / 0.2 / 0.5	„ ƒ …. 20- ƒ ƒ …
S7	PAUSE / RESUME	†	3 / 5 / 1.0 / 1.5	PAUSE ˆ ƒ ‹ , START ƒ €
S8	STEP	†	3 / 5 / 1.0 / 1.5	STEP ƒ event
S9	RESET	†	3 / 5 / 1.0 / 1.5	READY , ƒ , event 0
S10	‹ ƒ		25 / 5 / 1.0 / 1.5	ˆ ˆ€, ƒ ƒ …
S11	(ˆ)		abc / 5 / 1.0 / 1.5	ˆ ˆ€, ƒ ƒ …

• „ **9. Core (UI-) ˆ „ ‹ ‰ …ˆ**

ID		• • („†)	€ „	•„	„	TA „ „	TA „	
S1	3 / 5 / 1.0 / 1.5	20	11	11	15	1	0	3
S2	1 / 10 / 0.3 / 1.5	20	11	434	13	1	0	1
S3	10 / 3 / 1.5 / 0.5	20	20	0	22	11	10	2
S4	0 / 5 / 0.5 / 1.0	15	0	120	0	0	0	0
S5	1 / 1 / 2.0 / 0.5	20	6	0	6	6	6	1
S6	20 / 50 / 0.2 / 0.5	20	33	2451	54	1	0	20

• „ **10. UI ‰ „ ‹ core- ‹ ‰ ’•**

ID	Core: € „	UI: € „	Core: ‹ •„	UI: ‹ •„	UI- € „ / Core
S1	11	9	11	12	82%
S2	11	1	434	78	9%
S3	20	21	0	0	•
S4	0	0	120	64	•
S5	6	6	0	0	100%
S6	33	1	2451	59	3%

• „ **11. UI „ • †** • ‰ ‰ … ˆ

ID	• … ˆ „ ‰ • /	• ‰ „	
S1	, ƒ ƒ ; 20 ˆ	, ƒ ƒ TA . 20 ˆ - … … Œ 9, 12, 3, TA HELPING	
S2	20 ˆ	78, 1 (1), … Œ 1 (UI ƒ)	
S3	20 ˆ	–… Œ 21, 0	
S4	15 ˆ	–… Œ 0, 64, TA SLEEPING, 0 / 0 seats	
S5	20 ˆ	1 ’ … Œ … TA ƒ . –… Œ 6, 0	
S6	20 ˆ	RUNNING, …. 20 / 20, 59, … Œ 1 (UI ƒ)	
S7	PAUSE 4 ˆ	PAUSED ˆ event 15 › 15 (Œ …). START- … 30 ƒ €	
S8	PAUSE, STEP ˜ 5	Event 6 › 7 › 8 › 9 › 10 › 10. ‰ … STEP- "WAITING FOR EVENT"	
S9	10 ˆ , RESET	RESET- … : … Œ 5, 5, event 27. RESET- … : READY, event 0	
S10	START	INVALID INPUT. "‰ 0-20 ƒ ƒ … ."	
S11	START	INVALID INPUT. "'‰ ' Ž ƒ …ƒ “ … (: 'abc')."	

Synchronization- € ƒ ƒ ˆ :

- … mutex- ƒ ƒ , ƒ Œ … (Œ 20 , 50 ’ ƒ … ƒ 20).
- TA ‹ ƒ ’ … Œ .
- FIFO € : ƒ € ’ ƒ € … Œ .
- Semaphore- ƒ ƒ ‹ € €, TA ’ … ˆ €, ƒ ’ .
- Deadlock thread € … (ƒ reset ƒ … … Œ thread …).
- ‰ 0 ˆ ƒ Œ … Œ …, ƒ … ˆ ƒ Œ ….

10.3 „ ‰“Œ

ƒ ƒ ‹ . , Ž core- … synchronization- ‹ … ….

- **UI-** „ € † ‰ • ‰. • ’ ƒ UI Ž event- … ƒ ƒ • … Œ • core- ƒ (S2: core 11, UI 1; S6: core 33, UI 1). Core- … ‹ .

- **STEP Š€.** Event ”ˆ ƒ ˆ core- € … STEP- event € ƒ ƒ € .
- **reset(new_config).** ƒ … ƒ ƒ stop() ƒ … €, Ž ƒ ‹ ƒ . UI ƒ Ž Œ ƒ Œ ƒ ƒ ….
- **Seed.** UI- seed …ƒ … ˆ … € ƒ € …. Seed Ž ˆ € .
- **€ .** macOS ƒ … . UI ˆ … ‘ ‹ ƒ macOS- € .

10.4 ” „ ˆ ‰ €

- Event ”ˆ ƒ ˆ frame- event ƒ ƒ ƒ €, UI- core- … … ƒ.
- `reset-` ƒ ƒ Ž `stop()` ƒ …ƒ.
- UI- seed €, ƒ € … ƒ.
- ˆ ” … ƒ €, € ƒ.
- ž … … ˆ (Windows, Linux) ˆ , UI- ƒ.

10.5 ••„

thread, mutex, semaphore- … … … ƒ ƒ, thread- … € ƒ ‹ Œ ƒ € €, Ž ƒ ƒ … €, … ˆ € . Core- … synchronization ƒ ƒ ‹ € ƒ ‹ Ž Œ … ƒ … . , … ƒŒ ‹ Œ …€ ƒ € ….

f „ … †

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